

RAJAT BUTOLA (傅能杰)

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[My Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

I am a hands-on **Full-Stack AI Engineer** with expertise in designing, building, and deploying **production-grade AI applications**. I am a Ph.D. in **Electrical Engineering and Computer Science** from NYCU, Hsinchu, Taiwan. I build **on-premise** as well as **Cloud (Azure)** based **AI pipelines** and **Agentic** workflows that integrate domain-specific knowledge with LLM inference without routing sensitive data through external APIs that operate under real institutional constraints.

CORE COMPETENCIES & TECHNICAL SKILLS

AI / LLM / GenAI	Azure, OpenAI API, Open source LLMs, Agentic Workflows, LangChain, HuggingFace Transformers, Llama, RAG Pipelines, MCP, Prompt Engineering, Embedding models, Quantization
ML Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, n8n Workflow Automation
ML Techniques	Supervised / Unsupervised / Semi-supervised Learning, ANN, CNN, GNN, LSTM, Transformer, Autoencoder, YOLO, VGG, ResNet, XGBoost, Random Forest, SVM
Data & Systems	PostgreSQL, Time-series Modeling, Feature Engineering, Statistical modeling, Outlier Detection, Dimensionality Reduction, FAISS, Qdrant
Languages & Tools	Docker, Python, R, MATLAB, SQL, Pandas, NumPy, FastAPI, CUDA, NextJS, Node.js, Git, Jira
Semiconductor / EDA	Compact Modeling, Process Variation Analysis, TCAD, HSPICE, GAA/FinFET Device Simulation

PROFESSIONAL EXPERIENCE

AI Engineer | Tzu Chi Hospital | Research Dept., Taipei, Taiwan

Dec 2025 – Present

- Architected and developed a **Full Stack** production-grade AI platform for **Evidence-Based Medicine (EBM)** that assists clinicians in rapidly identifying, critically appraising, and synthesizing medical evidence for point-of-care decision support while also supporting end-to-end systematic review workflows at the same time maintains full methodological transparency.
- Designed a modular full-stack architecture using **FastAPI, Next.js (App Router), TypeScript, PostgreSQL, Redis, Celery, and Azure**, evolving the platform from an earlier Streamlit/SQLite prototype into a multi-user production system with role-based collaboration, secure authentication, and scalable deployment.
- Built **Agentic workflows** that integrate domain-specific knowledge with LLM inference without routing sensitive data through external APIs that operate under real institutional constraints.
- Shipped and maintained the multi-tenant platform with **RBAC**, PDF ingestion (**pdfplumber**), **multilingual (English/Chinese) UI**, and an AI-assisted clinical grading system all in production on **Azure** using **Nginx, systemd**, and **Let's Encrypt SSL** including secure authentication.
- Built an abstraction layer for LLM inference supporting both **local GGUF models** and **Azure OpenAI** deployments, enabling runtime model switching without application changes and allowing the platform to balance deployment flexibility, cost, and data privacy requirements.
- Designed an AI-driven literature **retrieval pipeline** that converts **natural-language** clinical questions into structured typical medical representations, performs hierarchical evidence retrieval.
- Engineered long-running AI workflows using **Server-Sent Events (SSE)** to stream literature retrieval and evidence analysis progress to the frontend, enabling responsive user interaction without the operational complexity of persistent WebSocket infrastructure.
- Designed the platform as a **clinical decision-support system rather than an autonomous decision-making system**, incorporating evidence traceability, deterministic validation, explicit confidence reporting, and human oversight throughout the clinical reasoning pipeline.

Post-Doctoral Researcher | Academia Sinica, Taipei, Taiwan

Feb 2025 – Nov 2025

- Designed and automated end-to-end genomic data pipelines (QC → normalization → feature engineering → modeling), reducing processing time by ~50% and enabling reliable ML predictions under high-dimensional, low-sample constraints.
- Developed machine learning models (logistic regression, random forest, SVM) for cancer treatment response prediction from RNA-seq data, achieving AUC ~0.75.
- Deployed logistic regression models (glmnet) via an interactive R Shiny application, enabling user-driven inference on gene expression data, containerized with Docker for reproducible deployment.

EDUCATION

Ph.D., Electrical Engineering & Computer Science (EECS) | National Yang-Ming Chiao Tung University, Taiwan
Sept 2018 – Dec 2023

Research: Deep Learning for Emerging GAA MOSFET Device and circuit Modeling

- Conducted research and analyzed large complex non-linear dataset of semiconductor nano-devices such as Nanosheet, CFET to extract valuable insights and ensuring tape-out readiness.
- Developed ANN/LSTM predictive models for process variation effects (gate length, oxide thickness, work-function fluctuation) on device DC/AC/RF electrical characteristics, reducing TCAD simulation time by >40%.
- Applied CNN and dimensionality reduction (DR) to identify problematic OPC mask patterns, accelerating lithography correction workflows while maintaining prediction accuracy.
- Developed compact neural network models as drop-in replacements for TCAD/HSPICE simulators, co-designed hybrid TCAD+SPICE datasets for semi-supervised training to minimize labeled data requirements.
- Collaborated with cross-functional teams and published work in International Journals (IEEE-TED, IEEE Transactions on Microwave Theory & Techniques, Elsevier, IEEE Access) and conferences.

Master, Electronics & Communication Engineering | Jaypee Institute of Information Technology University, India
Aug 2010 – May 2012

- Dissertation: Image Enhancement using Thresholding & Histogram Equalization (HE), built full evaluation pipeline for brightness-preserving HE frameworks using MATLAB and Python.

PUBLICATIONS

Journals:

1. **Rajat Butola**, Yuan S, Shieh GS. "ImmunoResponse Predictor: a GUI for accurate response prediction to immunotherapy," Bioinformatics and Genomics, PeerJ. Vol. 14, Aug, 2026, doi: 10.7717/peerj.21553.
2. **Rajat Butola** et al. "Compact Model Build upon Dynamic Weighting Artificial Neural Network Approach for Complementary Field Effect Transistors," IEEE Transactions on Electronic Devices (IEEE-TED), vol. 71, Jan. 2024.
3. **Rajat Butola** et al. "Application of Long Short-Term Memory Modeling Technique to Predict Process Variation Effects of Stacked Gate-All-Around Si Nanosheet Complementary-Field Effect Transistors," Computers & Electrical Engineering, Elsevier, vol. 105, Jan. 2023.
4. **Rajat Butola** et al. "Artificial Neural Network-Based Modeling for Estimating the Effects of Various Random Fluctuations on DC/Analog/RF Characteristics of GAA Si Nanosheet FETs," IEEE Trans. Microwave Theory & Techniques, vol. 70, no. 11, Nov. 2022.
5. **Rajat Butola** et al. "A Machine Learning Approach to Modeling Intrinsic Parameter Fluctuation of GAA Si Nanosheet MOSFETs," IEEE Access, vol. 10, July 2022.
6. **Rajat Butola** et al. "A Comprehensive Technique based on Machine Learning for Device and Circuit Modeling of Gate-All-Around Nanosheet Transistors," IEEE OJ-Nano, vol. 4, Oct. 2023.
7. S. R. Kola, Y. Li and **Rajat Butola**, "Statistical Device Simulation and Machine Learning of Process Variation Effects of Vertically Stacked Gate-All-Around Si Nanosheet CFETs," in IEEE Transactions on Nanotechnology, vol. 23, April, 2024.
8. P. Parashar, C. Akbar, T. S. Rawat, S. Pratik, **Rajat Butola**, S. H. Chen, Y-S Chang, S. Nuannimnoi, and A. S. Lin "Intelligent Photolithography Corrections Using Dimensionality Reductions," IEEE Photonics Journal, vol. 11, no. 5, Oct. 2019.
9. P. Parashar, C. H. Chen, C. Akbar, S. M. Fu, T. S. Rawat, S. Pratik, **Rajat Butola**, S. H. Chen, A. S. Lin. "Analytics-statistics mixed training and its fitness to semi-supervised manufacturing" PLoS ONE Journal, 2019.

Conferences:

1. **Rajat Butola** et al. "A DNN-Based Compact Modeling Technique for GAA Si NS FETs and Its Application in CMOS Circuit Simulation," International conferences on the science and technology of devices and materials (SSDM), Japan, 2023.
2. **Rajat Butola** et al. "Estimating the Process Variation Effects of Stacked Gate All Around Si Nanosheet CFETs Using Artificial Neural Network Modeling Framework," IEEE International Conference on Nanotechnology (IEEE-NANO), Spain, 2022.
3. **Rajat Butola** et al. "Deep Learning Approach to Estimating Work Function Fluctuation of Gate-All-Around Silicon Nanosheet MOSFETs with A Ferroelectric HZO Layer," IEEE-EDTM, Japan, 2022.
4. **Rajat Butola** et al. "Machine Learning Approach to Characteristic Fluctuation of Bulk FinFETs Induced by Random Interface Traps," IEEE-ISQED, USA, 2022.
5. **Rajat Butola** et al. "Deep Learning Approach to Modeling and Exploring Random Sources of Gate-All-Around Silicon Nanosheet MOSFETs," International Symposium on VLSI Technology, Systems and Applications (VLSI-TSA), Hsinchu, Taiwan, 2022.